Status of the Gunter & Scott 1990 Development in Lean

`main` at `fc313e9`. Build 1372 jobs, 0 errors, 3 `sorry` warnings, 0 other warnings. 124 modules, 46,442 lines, 2,122 theorems.

1. Extractions — what the paper contains

#	Kind	Count
1	Definitions (unnumbered)	11
2	Numbered theorems	16
3	Numbered lemmas	13
4	Numbered propositions	1
5	— numbered results, total	**30**
6	— conjuncts those 30 decompose into	93
7	Prose statements	146
8	Examples	1
9	Figures	4

2. Represented in Lean

#	Kind	Represented	Of	Not stated
1	Definitions	11	11	0
2	Theorems	16	16	0
3	Lemmas	13	13	0
4	Propositions	1	1	0
5	Prose statements	86	146	60
6	Examples	1	1	0

3. Corrected — could not be transcribed as printed

#	Kind	Printing	Reading	Semantic	Corrected	— proven	— unproven
1	Definitions	1	0	1	2	2	0
2	Theorems	1	1	2	4	2	2
3	Lemmas	1	2	0	3	2	1
4	Propositions	0	0	0	0	0	0
5	Examples	0	0	0	0	0	0
6	— total	3	3	3	**9**	**6**	**3**
7	Prose statements	—	—	—	—	—	not classified

Printing error — the page is garbled; the mathematics is right

#	Where	What	Corrected to	Proven
1	Lemma 9	the PDF drops every ⊗ and every ⊥, so which operators the laws range over is unreadable	`lem9_1`…`lem9_6`; items 3 and 5 also carry kernel-checked negations of the printed forms	**yes**
2	Theorem 14	the list of characterizations is garbled	`thm14`, both directions	**yes**
3	§7.4 worked example	reverses the paper's own definition — printed `b ⊢ a` where both papers give `a ⊢ b`	read in the paper's direction; the paper's counts 1, 2, 5, 20 select it over the rival	**yes**

Reading error — ours; the paper is right

#	Where	What	Corrected to	Proven
1	Theorem 29, sentence 2	dropped “domain” from “any bifinite domain”, making it false	`Thm29SecondAtDomains`	no
2	Lemma 28	quantified over all `U`; the paper states it over its own `U`	`PRep.Lemma28AtU`, nine conjuncts, no hypotheses	**yes**
3	Lemma 30	same shape — a universal closure the paper does not claim	`Lemma30AtV`, arity 1	no

Semantic error — the mathematics is wrong as stated

#	Where	Whose	What	Corrected to	Proven
1	Theorem 26	**paper**	false for any signature with two or more 0-ary slots, including the paper's own `(2,0,0,0,0,0)`	`thm26` with `hs : ∀ i, 0 < s i`	**yes**
2	Theorem 7	ours	guarded on compactness where the sentence is about boundedness; on the paper's index the join always exists, so it tests neither	`StepFunctionsDecidable` over `IsStepEnumeration`, existentially quantified	no
3	§7.4 order relation	**paper**	not reflexive — `b = (⊥,∅)` fails `b ⊢ b`; it is the strict part	reflexive closure; `Colimit.V` built on it, `isoPlus : V ≃o Plus V`	**yes**

4. Proof state

#	Kind	Proven	Unproven	Of
1	Theorems	14	2	16
2	Lemmas	12	1	13
3	Propositions	1	0	1
4	— numbered results	**27**	**3**	30
5	Prose statements	70	16	86 stated
6	Examples	1	0	1

#	Unproven result	State
1	Theorem 7	sentences 1 and 2 proven; the recursive forms are not
2	Theorem 29	sentence 1 proven; sentence 2 is not
3	Lemma 30	stated at `V`, arity 1 — one named input left

All three reduce to **three** statements, each carried by Gunter & Scott in 1990. Each is a Lean `theorem` ending in `sorry`, so the build reports it (r0052):

#	The unproven statement	Lean theorem	Blocks
1	`RecursiveNormal` for `K(D → E)` — decide whether two basis elements are bounded, and compute their join's index	`R49.Agent3.scottHomCRecursive_unproven`	Theorem 7, arrow
2	the same over `K(D ⊸ E)` — no proved reduction carries recursiveness across `K(D ⊸ E) ↪ K(D → E)`	`R49.Agent3.strictHomCRecursive_unproven`	Theorem 7, strict
3	`Theorem29Normal` at infinite `K(E)` — realize the type over the tower's image, not the η-image; finite bases are proven	`LemThirty.theorem29Normal_unproven`	Theorem 29, Lemma 30

5. Refuted — a statement proved false, with a kernel-checked ¬

#	Kind	Count
1	the paper's own statement	1
2	our transcription of it	4
3	— total	**5**

#	Statement refuted	Whose	Refutation	Witness
1	Theorem 26 as printed	**paper**	`not_thm26Printed_of_two_zero_arities`	any signature with two 0-ary slots, including the paper's own `(2,0,0,0,0,0)`
2	Theorem 29 sentence 2, without the word "domain"	ours	`not_thm29Second`	`E := Flat (Set ℕ)`
3	`Theorem29Normal` with `[Domain E]` deleted	ours	`not_thm29NormalWithoutDomain`	`A∞` is countable, `Flat (Set ℕ)` is bifinite with uncountable basis
4	Lemma 28 with the carrier quantified	ours	`not_forall_lemma28`, and `not_forall_lemma28_bcd` shows no instance binder rescues it	`Flat Empty`
5	Lemma 30 with the carrier quantified	ours	`not_forall_lemma30`	`Flat Empty`, via Lemma 28's conjuncts

Rows 2–5 are our own over-general transcriptions, and each has a corrected statement that is not refuted: `Thm29SecondAtDomains`, `Theorem29Normal`, `PRep.Lemma28AtU` (**proved**), `Lemma30AtV`. Only row 1 convicts the paper.

6. Proof work state

#	Kind	Left with a `sorry`
1	Definitions	0
2	Theorems	**3**
3	Lemmas	0
4	Propositions	0
5	Prose statements	0
6	Examples	0

`sorry` **3**, `axiom` 0, since r0052; `sorry` was 0 from r0042 to r0051. Every unproved result is a `theorem … := sorry`, so the unproven count **is** the `sorry` count: the three of table 4's second table above. Exactly three package constants depend on `sorryAx` — those three theorems — and no other declaration does, because none applies them: every consumer still takes its claim as an explicit hypothesis.

Reproducing

```
scripts/counts.sh                     # modules, lines, theorems, sorry
scripts/numbered-status.sh            # declarations per numbered result
scripts/a6-env-scan.sh <out>          # then a6-summarize.py
scripts/a8-claim-check.sh             # prose staleness against the environment
scripts/compile.sh -r <round>         # build
```

Tables 1 and 9 come from `pdftotext -layout` on the paper. The text layer drops the `fi` ligature — `Definition` reads `De nition`, `bifinite` reads `bi nite` — so a grep for `Definition` returns 0 and is wrong.